9.9 Solving Quadratics Using the Quadratic Formula

Lesson Introduction

Benchmark	MA.912.AR.3.1 Given a mathematical or real-world context, write, and solve one-variable quadratics equations over the real number system.	Learning Target	• I can solve a quadratic using the quadratic formula.

Benchmark Coherence	Building On	Working Toward
	MA.8.AR.2.3 Given an equation in the form of $x^2 = p$ and $x^3 = q$, where p is a whole number and q is an integer, determine the real solutions.	MA.912.AR.3.2 Given a mathematical or real-world context, write and solve one-variable quadratic equations over the real and complex number systems.

Essential Questions	• How can I solve a quadratic equation using the quadratic formula? • What situations are best served by using the quadratic formula?
Lesson Narrative	In this lesson, students are introduced to the quadratic formula. Students will use the quadratic formula to solve quadratic equations. In addition to using the quadratic formula, students will learn how the quadratic formula is derived.

Component Summary				
	9.9.1 Warm-Up (~5 min.)	"Which one doesn't belong?" activity *(SE p. 103)*	9.9.3 Try It (10 min.)	Practice using the quadratic formula to find the solutions to a quadratic equation *(SE p. 106)*
	9.9.2 Guided Instruction (15 min.)	Introduction to the use of the quadratic formula to find the solutions to a quadratic equation *(SE p. 104)*	9.9.4 Your Turn! (10 min.)	Independent practice on using the quadratic formula to find the solutions to a quadratic equation *(SE p. 106)*
	9.9.5 Wrap-Up (~5 min.)	Self-reflection learning period regarding the content of this lesson *(SE p. 107)*		

Resources	Glossary	Materials	Additional Preparation
	<u>Key Terms:</u> • Quadratic formula • Nonreal value <u>Additional Terms:</u> N/A	• N/A	• Work through each question prior to instruction. There are many opportunities for miscalculations in the process. Additionally, work through the derivation of the quadratic formula prior to instruction. Students may struggle with this section and may need additional explanations.

<table>
<tr><td rowspan="4">Teacher Tips</td><td>Common Misconceptions</td><td>Things to Consider</td></tr>
<tr><td>Students may misidentify the a, b, and c terms.
Students may forget to use the opposite of the b term in the formula.
Students may make calculation errors when evaluating.
Students may forget to create two solutions at the ± part of the process.</td><td>This lesson walks students through how the quadratic equation was derived. This provides a solid conceptual understanding for students to make the connections to what they have been learning about quadratics. The primary purpose of the targeted standard for the lesson is writing and solving quadratic equations.</td></tr>
<tr><td>Literacy and Language Routines</td><td>Mathematical Thinking and Reasoning (MTR) Standard</td></tr>
<tr><td>Notice and Wonder
The Warm-Up activity in this lesson provides students with four parabolas to examine. Consider applying “Notice and Wonder” routines on each parabola, positioning students to discover the comparisons between key features of each graph.</td><td>MA.K12.MTR.3.1: Mathematical Fluency
The process of using the quadratic formula can result in several steps that require substantial calculations. It will be essential for students to maintain accuracy when completing each step and maintain flexibility depending on the given situation.</td></tr>
<tr><td>Differentiate Instruction</td><td colspan="2">9.9.2 Guided Instruction provides students with an opportunity for solving quadratic equations using the quadratic formula. Students will be able to show mastery of multiple components of the question, allowing for more easily identifiable differentiation opportunities. Consider creating a visual aid, such as an anchor chart, that shows the process for the quadratic formula to provide a support for visual learners. Use a different color for the a, b, and c values so visual learners can connect where the information came from.

The 9.9.4 “Your Turn” section provides an opportunity for students to solve a quadratic equation using the quadratic formula. Student performance on these two questions can provide valuable data for differentiation.</td></tr>
<tr><td colspan="3">Lesson Notes:</td></tr>
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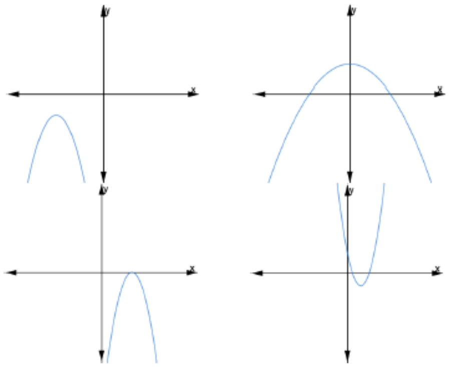
9.9.1 Warm-Up: Which One Doesn't Belong?

Component Overview: *Students will review four parabolas to determine which doesn't belong using their knowledge of key features.*

9.9.1 Warm-Up: Which One Doesn't Belong?



1. Four different parabolas are shown. Choose one graph that does not belong and explain your choice using key features.
   Answers will vary.
   - *The bottom right graph does not belong because it is opening upward (has a positive a-value).*
   - *The top left graph does not belong because it does not touch the x-axis (has no real solutions).*
   - *The top right graph does not belong because it opens wider than the other three or because it has a positive and negative solution.*
   - *The bottom left graph does not belong because it has only one solution.*

Challenge students to explain how the one that doesn't belong could be modified to fit the others. Have them explain their answer using appropriate mathematical terminology they have learned during this and prior units.

- *What key features did you choose to look for first?*
- *What process did you go through to compare key features of these graphs?*

9.9.2 Guided Instruction: Quadratic Formula

Component Overview: *Students will receive guided instruction on the use of the quadratic formula to find the solutions to a quadratic equation.*

9.9.2 Guided Instruction: Quadratic Formula

1. Rewrite $x = -5 \pm \sqrt{81}$ as two separate values.
   $x = -5 + \sqrt{81} = 4$
   $x = -5 - \sqrt{81} = -14$

2. Determine the two different values $x = -b \pm \sqrt{b^2 - 4ac}$ given $a = -2$, $b = -15$, and $c = -6$. Leave in square root form.
   $x = -(-15) \pm \sqrt{(-15)^2 - 4(-2)(-6)}$
   $x = 15 \pm \sqrt{225 - 48}$
   $x = 15 \pm \sqrt{177}$
   $x = 15 + \sqrt{177}$ $\quad$ $x = 15 - \sqrt{177}$

3. Determine the two different values $x = -b \pm \sqrt{b^2 - 4ac}$ given $a = 1$, $b = 4$, and $c = 8$. Leave in square root form.
   $x = -4 \pm \sqrt{(4)^2 - 4(1)(8)}$
   $x = -4 \pm \sqrt{16 - 32}$
   $x = -4 \pm \sqrt{-16}$
   $x = -4 + \sqrt{-16}$ $\quad$ $x = -4 - \sqrt{-16}$

The square root of a negative number is a **non-real value**.

Solving quadratics when given standard form is similar to isolating the variable x from standard form, except that standard form is set equal to 0.

Standard Form Set Equal to 0	$ax^2 + bx + c = 0$
From standard form the vertex of the parabola can be found using $\left(\frac{-b}{2a}, \frac{c}{a} - \frac{b^2}{4a^2}\right)$. Rewrite standard form to vertex form using this vertex.	$a(x-h)^2 + k = 0$ $\left(x + \frac{b}{2a}\right)^2 + \frac{c}{a} - \frac{b^2}{4a^2} = 0$
Subtract $\frac{c}{a}$ from both sides.	$\left(x + \frac{b}{2a}\right)^2 - \frac{b^2}{4a^2} = -\frac{c}{a}$

- *What part of the equation signifies that there will be two separate solutions?*
- *Under what circumstances do you think there might be a different number of solutions?*

In question 2, students may struggle with inputting values and evaluating. Students who struggle with either one at this stage will have challenges during the quadratic formula process. Use performance data from this question for potential differentiation.

Share the concept of "non-real," but don't go too in depth with describing this concept to the point of imaginary numbers. Students should be aware that non-real solutions would arise from this scenario, but they should not be expected to describe or define what they are. Students should be encouraged to relate this concept in the algebraic representation to their prior explorations of graphs with no real solution.

The conversation of this part of the lesson is to derive the quadratic formula. Don't focus on memorization of steps, but rather conceptual understanding of how this connects to their work with quadratics thus far. Students should understand the purpose of each step and how to successfully complete them with instructional support. Students may recognize the steps from lesson 8.

9.9.2 Guided Instruction: Quadratic Formula (continued)

Add the opposite of $-\frac{b^2}{4a^2}$ to both sides.	$\left(x+\frac{b}{2a}\right)^2=-\frac{c}{a}+\frac{b^2}{4a^2}$
In this step, a mathematician provided an equivalent form. The algebra that the mathematician used will be learned in Algebra 2. All of the steps the mathematician used are not shown.	$\left(x+\frac{b}{2a}\right)^2=-\frac{c}{a}+\frac{b^2}{4a^2}$ $\left(x+\frac{b}{2a}\right)^2=\frac{b^2-4ac}{4a^2}$
Take the square root of both sides. When taking the square root of both sides include $\pm$ to represent both of the possible outputs.	$\left(x+\frac{b}{2a}\right)^2=\frac{b^2-4ac}{4a^2}$ $\sqrt{\left(x+\frac{b}{2a}\right)^2}=\pm\sqrt{\frac{b^2-4ac}{4a^2}}$ $x+\frac{b}{2a}=\pm\sqrt{\frac{b^2-4ac}{4a^2}}$
Subtract $\frac{b}{2a}$ from both sides.	$x=-\frac{b}{2a}\pm\sqrt{\frac{b^2-4ac}{4a^2}}$
In this step, a mathematician provided an equivalent form. The algebra that the mathematician used will be learned in Algebra 2. All of the steps the mathematician used are not shown.	$x=\frac{-b}{2a}\pm\sqrt{\frac{b^2-4ac}{4a^2}}$ $x=\frac{-b\pm\sqrt{b^2-4ac}}{2a}$

By isolating the variable x, a formula for solving any quadratic in standard form that is set equal to 0 was created. This formula is called the quadratic formula.

Quadratic Formula

$$x=\frac{-b\pm\sqrt{b^2-4ac}}{2a}$$

The quadratic formula is used to solve a quadratic in standard form that is set equal to 0, $ax^2+bx+c=0$.

Why would there be a ± sign when you take the square root of both sides?

Challenge students to work through the steps with their partners.

Some students may struggle with completing these steps without a numeric value. Consider working with these students on reviewing literal equations in linear and quadratic situations to strengthen those skills.

Consider creating a visual aid for the quadratic formula. This could include a detailed anchor chart or table-tent for students. Use different colors for the a, b, and c values so visual learners can connect where the input values came from. For auditory learners, be consistent with language and explain each step as you demonstrate it.

4. Complete the table.

Quadratic Equation	Quadratic Formula
$12x^2 + 8x - 15 = 0$ $a = 12$ $b = 8$ $c = -15$	$x = \frac{-8 \pm \sqrt{(8)^2 - 4(12)(-15)}}{2(12)}$ $x = \frac{-8 \pm \sqrt{64 + 720}}{24}$ $x = \frac{-8 \pm \sqrt{784}}{24}$ $x = \frac{-8 \pm 28}{24}$ $x = \frac{-8+28}{24}$ $x = \frac{-8-28}{24}$ **Solutions:** $x = \frac{5}{6}$ $x = -\frac{3}{2}$
$x^2 + 6x + 7 = -6$ $x^2 + 6x + 13 = 0$ $a = 1$ $b = 6$ $c = 13$	$x = \frac{-6 \pm \sqrt{(6)^2 - 4(1)(13)}}{2(1)}$ $x = \frac{-6 \pm \sqrt{36 - 52}}{2}$ $x = \frac{-6 \pm \sqrt{-16}}{2}$ **Solutions:** *Non-real solutions*
$-5x^2 - 5x + 18 = 4x + 17$ $-5x^2 - 9x + 1 = 0$ $a = -5$ $b = -9$ $c = 1$	$x = \frac{-(-9) \pm \sqrt{(-9)^2 - 4(-5)(1)}}{2(-5)}$ $x = \frac{-(-9) \pm \sqrt{81 + 20}}{-10}$ $x = \frac{9 \pm \sqrt{101}}{-10}$ $x = \frac{9+\sqrt{101}}{-10}$ $x = \frac{9-\sqrt{101}}{-10}$ **Solutions:** $x = \frac{9+\sqrt{101}}{-10}$ $x = \frac{9-\sqrt{101}}{-10}$

Focus on repetitive use of language with each step. Students should hear the appropriate mathematical language and explanation at each step so that they can follow along and apply it to their own questions.

9.9.3 Try It: Solving With the Quadratic Formula

Component Overview: *Students will practice using the quadratic formula to find the solutions to a quadratic equation. This activity is an opportunity for students to practice using the quadratic formula. Since this lesson has a "Try It" activity, students can work with a partner on this activity to practice what they've learned.*

9.9.3 Try It: Solving with the Quadratic Formula

1. Use the quadratic formula to solve $x^2 + 4x + 4 = 0$.

$a = 1$ $b = 4$ $c = 4$

$$x = \frac{-4 \pm \sqrt{(4)^2 - 4(1)(4)}}{2(1)}$$

$$x = \frac{-4 \pm \sqrt{16 - 16}}{2}$$

$$x = \frac{-4 \pm \sqrt{0}}{2}$$

$$x = -\frac{4}{2}$$

$$x = -2$$

What does the graph look like when there is only one solution?

Watch for how students begin these questions. If they aren't writing down the values of a, b, and c and are making errors, consider asking them to do that. Reiterate process language like starting steps regularly through instruction and when working with individuals or groups of students.

2. Use the quadratic formula to solve $3x^2 - 7x - 3 = 3x + 8$.

$3x^2 - 10x - 11 = 0$

$a = 3$ $b = -10$ $c = -11$

$$x = \frac{-(-10) \pm \sqrt{(-10)^2 - 4(3)(-11)}}{2(3)}$$

$$x = \frac{10 \pm \sqrt{100 + 132}}{6}$$

$$x = \frac{10 \pm \sqrt{232}}{6}$$

$$x = \frac{10 \pm 2\sqrt{58}}{6}$$

$x = \frac{10+2\sqrt{58}}{6}$ $\quad x = \frac{10-2\sqrt{58}}{6}$

$x = \frac{5+\sqrt{58}}{3}$ $\quad x = \frac{5-\sqrt{58}}{3}$

9.9.4 Your Turn!

Component Overview: *Students will independently practice using the quadratic formula to find the solutions to a quadratic equation. This activity is designed for students to practice using the quadratic equation on their own.*

9.9.4 Your Turn!

1. Solve $-7x^2 + 10x - 1 = 0$.

$a = -7\ b = 10\ c = -1$

$$x = \frac{-(10) \pm \sqrt{(10)^2 - 4(-7)(-1)}}{2(-7)}$$

$$x = \frac{-(10) \pm \sqrt{100 - 28}}{-14}$$

$$x = \frac{-(10) \pm \sqrt{72}}{-14}$$

$$x = \frac{-(10) \pm 6\sqrt{2}}{-14}$$

$x = \frac{-(10)+6\sqrt{2}}{-14}$ $\quad$ $x = \frac{-(10)-6\sqrt{2}}{-14}$

$x = \frac{5-3\sqrt{2}}{7}$ $\quad$ $x = \frac{5+3\sqrt{2}}{7}$

2. Determine the solution(s) for $5x^2 - 7x + 5 = 2x^2 - 17x$.

$3x^2 + 10x + 5 = 0$

$a = 3\ b = 10\ c = 5$

$$x = \frac{-(10) \pm \sqrt{(10)^2 - 4(3)(5)}}{2(3)}$$

$$x = \frac{-10 \pm \sqrt{100 - 60}}{6}$$

$$x = \frac{-10 \pm \sqrt{40}}{6}$$

$$x = \frac{-10 \pm 2\sqrt{10}}{6}$$

$x = \frac{-10+2\sqrt{10}}{6}$ $\quad$ $x = \frac{-10-2\sqrt{10}}{6}$

$x = \frac{-5+\sqrt{10}}{3}$ $\quad$ $x = \frac{-5-\sqrt{10}}{3}$

Consider differentiating this activity based on performance in earlier parts of this lesson. For example, for students who are generally struggling with getting started or the basic components of the quadratic formula, allow them more time to work on question 1 only and get to question 2 if time permits. This group may need additional support, while other students can complete the first question on their own and may only need support in the second question.

Students may struggle with question 2. Listen for student talk about identifying a, b, and c before setting it equal to 0.

What are some of the differences in how you solved question 1 versus how you solved question 2? What advice would you give another student attempting to solve both questions?

9.9.5 Wrap-Up: Reflection

Component Overview: *Students will complete a self-reflection learning pyramid regarding the content of this lesson. This activity should be done independently so that each student can complete their own personal reflection.*

9.9.5 Wrap-Up: Reflection

1. Complete the learning pyramid.
   Answers will vary.

One question I would like to have answered ...

One mistake I learned from today ...

One thing I am not sure about ... YET!

One thing I am confident I know is ...

Encourage students to be honest when completing this learning pyramid. Their responses provide valuable information on what students feel confident and don't feel confident in.